

PROBLEM 5.3

5.3 Classify the following processes of a closed system as possible, impossible, or indeterminate.

	Entropy Change	Entropy Transfer	Entropy Production
(a)	>0	0	
(b)	<0		>0
(c)	0	>0	
(d)	>0	>0	
(e)	0	<0	
(f)	>0		<0
(g)	<0	<0	

Eg. 5.2 for a closed system:

$$[\text{Entropy Change}] = [\text{Entropy Transfer}] + [\text{Entropy Production}]$$

(a) $[>0] = [0] + [\text{Entropy Production}] \Rightarrow [\text{Entropy Production}] > 0$. Possible.

(b) $[\text{Entropy Production}] > 0$. Possible.

(c) $[0] = [>0] + [\text{Entropy Production}] \Rightarrow [\text{Entropy Production}] < 0$. Impossible.

(d) $[>0] = [>0] + [\text{Entropy Production}]$. No conclusion about $[\text{Entropy Production}]$ can be reached without actual values for the other terms. Indeterminate.

(e) $[0] = [<0] + [\text{Entropy Production}] \Rightarrow [\text{Entropy Production}] > 0$. Possible.

(f) $[\text{Entropy Production}] < 0$. Impossible.

(g) $[<0] = [<0] + [\text{Entropy Production}]$. No conclusion about $[\text{Entropy Production}]$ can be reached without actual values for the other terms. Indeterminate.

PROBLEM 5.5

5.5 As shown in Fig. P5.5, a reversible power cycle R and an irreversible power cycle I operate between the same hot and cold thermal reservoirs. Cycle I has a thermal efficiency equal to one-third of the thermal efficiency of cycle R.

(a) If each cycle receives the same amount of energy by heat transfer from the hot reservoir, determine which cycle (i) develops greater net work, (ii) discharges greater energy by heat transfer to the cold reservoir.

(b) If each cycle develops the same net work, determine which cycle (i) receives greater energy by heat transfer from the hot reservoir, (ii) discharges greater energy by heat transfer to the cold reservoir.

SCHEMATIC & GIVEN DATA:

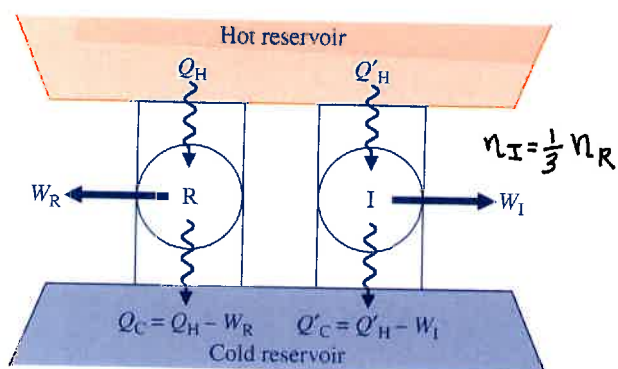


Fig. P5.5

$$n_R = \frac{W_R}{Q_H}, \quad n_I = \frac{W_I}{Q'_H}$$

ANALYSIS:

(a) $Q_H = Q'_H$

(i) $n_I = \frac{1}{3} n_R$

$$\frac{W_I}{Q_H} = \frac{1}{3} \frac{W_R}{Q_H} \Rightarrow W_I = \frac{1}{3} W_R \Rightarrow \text{cycle R develops greater net work.}$$

(ii) Energy balances give

$$\left. \begin{array}{l} Q_C = Q_H - W_R \\ Q'_C = Q'_H - W_I \end{array} \right\} \Rightarrow \begin{aligned} Q'_C - Q_C &= W_R - W_I \\ &= W_R - \frac{1}{3} W_R \\ &= \frac{2}{3} W_R \end{aligned}$$

$\Rightarrow$ Cycle I discharges greater energy by heat transfer to the cold reservoir

Summary: Cycle I develops less net work than cycle R but discharges greater heat transfer to the cold reservoir.

(b) $W_I = W_R$

(i) $n_I = \frac{1}{3} n_R$

$$\frac{W_R}{Q'_H} = \frac{1}{3} \frac{W_R}{Q_H} \Rightarrow Q'_H = 3 Q_H$$

Cycle I receives greater energy by heat transfer from the hot reservoir.

PROBLEM 5.5 (Continued)

(ii) Energy balances give

$$\left. \begin{aligned} Q_c &= Q_H - W_R \\ Q'_c &= Q'_H - W_R \end{aligned} \right\} \begin{aligned} Q'_c - Q_c &= Q'_H - Q_H \\ &= 3Q_H - Q_H \\ &= 2Q_H \end{aligned}$$

$\Rightarrow$ Cycle I transfers greater energy by heat transfer to the cold reservoir.

Summary: For equal net work developed, cycle I receives greater energy by heat transfer from the hot reservoir and discharges greater energy by heat transfer to the cold reservoir.

PROBLEM 5.10

5.10 Figure P5.10 shows two power cycles, denoted 1 and 2, operating in series, together with three thermal reservoirs. The energy transfer by heat into cycle 2 is equal in magnitude to the energy transfer by heat from cycle 1. All energy transfers are positive in the directions of the arrows.

(a) Determine an expression for the thermal efficiency of an overall cycle consisting of cycles 1 and 2 expressed in terms of their individual thermal efficiencies.

(b) If cycles 1 and 2 are each reversible, use the result of part (a) to obtain an expression for the thermal efficiency of the overall cycle in terms of the temperatures of the three reservoirs, T_H , T , and T_C , as required. Comment.

(c) If cycles 1 and 2 are each reversible and have the same thermal efficiency, obtain an expression for the intermediate temperature T in terms of T_H and T_C .

SCHEMATIC & GIVEN DATA

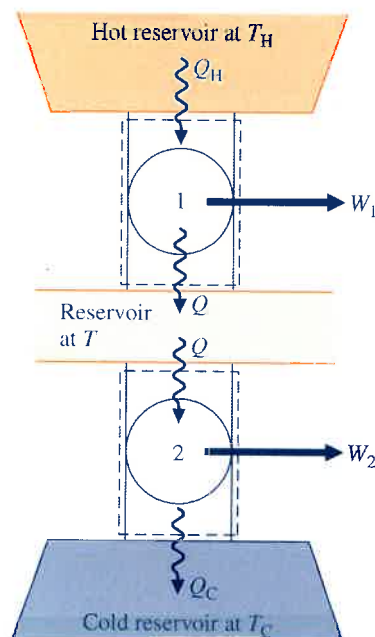


Fig.P5.10

KNOWN: Two power cycles operate in series.

FIND: For three scenarios obtain expressions for the thermal efficiency of the overall cycle formed from the two.

ENGINEERING MODEL:

1. The systems denoted by 1 and 2 in the figure each undergo power cycles.
2. There are three thermal reservoirs at temperatures T_H , T , and T_C , respectively.
3. All energy transfers are positive in the direction of the accompanying arrow.

ANALYSIS:

(a) For the overall cycle $(W_1 + W_2) = Q_H - Q_C$. Moreover,

$$\eta = \frac{W_1 + W_2}{Q_H} = 1 - \frac{Q_C}{Q_H}$$

For cycle 1,

$$\eta_1 = \frac{W_1}{Q_H} = 1 - \frac{Q}{Q_H} \Rightarrow Q_H = \frac{Q}{(1 - \eta_1)}$$

For cycle 2

$$\eta_2 = \frac{W_2}{Q} = 1 - \frac{Q_C}{Q} \Rightarrow Q_C = Q(1 - \eta_2)$$

Collecting results

$$\eta = 1 - (1 - \eta_1)(1 - \eta_2) = 1 - (1 - \eta_1 - \eta_2 + \eta_1 \eta_2)$$

$$= \eta_1 + \eta_2 - \eta_1 \eta_2$$

← (a)

①

(b) If cycles 1 and 2 are each reversible, then Eq. 5.9 gives

$$\eta_1 = 1 - \frac{T}{T_H}, \quad \eta_2 = 1 - \frac{T_C}{T}$$

Substituting into the result of part (a)

$$\eta = \left(1 - \frac{T}{T_H}\right) + \left(1 - \frac{T_C}{T}\right) - \left(1 - \frac{T}{T_H}\right)\left(1 - \frac{T_C}{T}\right)$$

PROBLEM 5.10 (Continued)

On reduction

$$\eta = 1 - \frac{T}{T_H} + 1 - \frac{T_C}{T} - \left[1 - \frac{T}{T_H} - \frac{T_C}{T} + \frac{T}{T_H} \frac{T_C}{T} \right]$$

$$\textcircled{2} \quad = 1 - \frac{T_C}{T_H} \quad \longleftarrow (b)$$

(c) If cycles 1 and 2 are each reversible

$$\eta_1 = 1 - \frac{T}{T_H}, \quad \eta_2 = 1 - \frac{T_C}{T}$$

If $\eta_1 = \eta_2$,

$$1 - \frac{T}{T_H} = 1 - \frac{T_C}{T}$$

$$\Rightarrow T^2 = T_C T_H$$

$$\textcircled{3} \quad \Rightarrow T = \sqrt{T_C T_H} \quad \longleftarrow (c)$$

1 Sample calculation: $\eta_1 = 0.25$, $\eta_2 = 0.40$

$$\eta = 0.25 + 0.40 - (0.25)(0.40) = 0.55$$

The combined cycle thermal efficiency is greater than that for each of individual cycles.

2. Referring again to Eq. 5.9, this result indicates that the combined cycle formed from two reversible cycles, as shown in the figure, is also reversible, as expected.

3 Sample calculation: $T_C = 360\text{ K}$, $T_H = 1000\text{ K}$

$$T = \sqrt{(360\text{ K})(1000\text{ K})} = 600\text{ K}$$

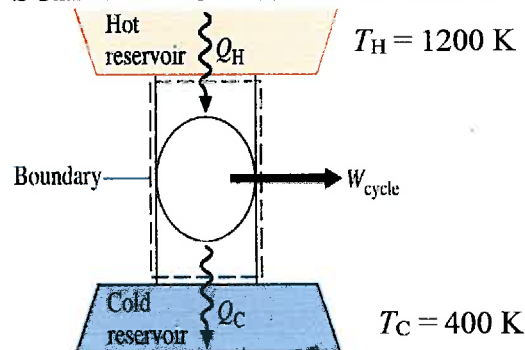
5.14 A power cycle receives energy Q_H by heat transfer from a hot reservoir at $T_H = 1200\text{ K}$ and rejects energy Q_C by heat transfer to a cold reservoir at $T_C = 400\text{ K}$. For each of the following cases, determine whether the cycle operates *reversibly*, operates *irreversibly*, or is *impossible*.

- (a) $Q_H = 900\text{ kJ}$, $W_{\text{cycle}} = 450\text{ kJ}$
- (b) $Q_H = 900\text{ kJ}$, $Q_C = 300\text{ kJ}$
- (c) $W_{\text{cycle}} = 600\text{ kJ}$, $Q_C = 400\text{ kJ}$
- (d) $\eta = 75\%$

KNOWN: Data are provided for a power cycle operating between hot and cold reservoirs with specified temperatures.

FIND: In each case, determine whether the cycle operates reversibly, operates irreversibly, or is impossible.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL: (1) The system is a closed system undergoing a cycle.

ANALYSIS: The maximum theoretical thermal efficiency for *any* power cycle operating between two reservoirs is given by Eq. 5.9. With the specified temperatures

$$\eta_{\max} = 1 - \frac{T_C}{T_H} = 1 - \frac{400\text{ K}}{1200\text{ K}} = 0.667 \text{ (66.7\%)}$$

(a) $Q_H = 900\text{ kJ}$, $W_{\text{cycle}} = 450\text{ kJ}$

$$\eta = \frac{W_{\text{cycle}}}{Q_H} = \frac{450\text{ kJ}}{900\text{ kJ}} = 0.5 \text{ (50\%)}$$

Since $\eta < \eta_{\max}$ the cycle operates irreversibly. (a)

(b) $Q_H = 900\text{ kJ}$, $Q_C = 300\text{ kJ}$

From an energy balance for the cycle: $W_{\text{cycle}} = Q_H - Q_C = 600\text{ kJ}$.

Thus

$$\eta = \frac{W_{\text{cycle}}}{Q_H} = \frac{600\text{ kJ}}{900\text{ kJ}} = 0.667 \text{ (66.7\%)}$$

Since $\eta = \eta_{\max}$ the cycle operates reversibly. (b)

(c) $W_{\text{cycle}} = 600\text{ kJ}$, $Q_C = 400\text{ kJ}$

From an energy balance for the cycle: $Q_H = W_{\text{cycle}} + Q_C = 1000\text{ kJ}$.

Thus

$$\eta = \frac{W_{\text{cycle}}}{Q_H} = \frac{600\text{ kJ}}{1000\text{ kJ}} = 0.6$$

Since $\eta < \eta_{\max}$ the cycle operates irreversibly. (c)

(d) $\eta = 75\%$

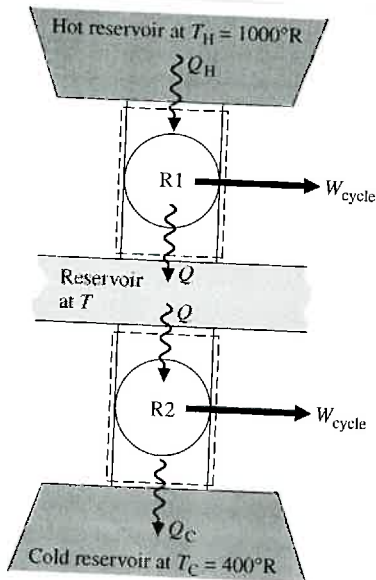
Since $\eta > \eta_{\max}$, the cycle cannot operate as claimed: the cycle is impossible. (d)

PROBLEM 5.22

KNOWN: Data are provided for two reversible power cycle in series that produce the same net work.

FIND: Determine (a) the intermediate temperature T and the thermal efficiency for each of the two power cycles. (b) For a single cycle, determine the thermal efficiency and net work.

SCHEMATIC & GIVEN DATA:



ANALYSIS:

(a) Energy balances: Since W_{cycle} is the net work developed by each cycle

$$W_{\text{cycle}} = Q_H - Q = Q - Q_C$$

$$\Rightarrow Q = \frac{Q_H + Q_C}{2} \quad (1)$$

Since the cycles are reversible, Eq. 5.7 applies to each:

$$\frac{Q}{Q_H} = \frac{T}{T_H}, \quad \frac{Q_C}{Q} = \frac{T}{T}$$

$$\Rightarrow Q_H = \frac{T_H}{T} Q$$

$$Q_C = \frac{T_C}{T} Q$$

Substituting these results into Eq. (1)

$$Q = \frac{\frac{T_H}{T} Q + \frac{T_C}{T} Q}{2}$$

$$\Rightarrow T = \frac{T_H + T_C}{2} = \frac{1000^\circ\text{R} + 400^\circ\text{R}}{2} = 700^\circ\text{R} \leftarrow$$

Then, for R1:

$$\eta_1 = 1 - \frac{T}{T_H} = 1 - \frac{700^\circ\text{R}}{1000^\circ\text{R}} = 0.3 \text{ (30\%)} \leftarrow$$

For R2:

$$\eta_2 = 1 - \frac{T_C}{T} = 1 - \frac{400^\circ\text{R}}{700^\circ\text{R}} = 0.43 \text{ (43\%)} \leftarrow$$

(b) For the single reversible cycle operating between $T_H = 1000^\circ\text{R}$ and $T_C = 400^\circ\text{R}$,

$$\textcircled{1} \quad \eta_{\text{max}} = 1 - \frac{T_C}{T_H} = 1 - \frac{400^\circ\text{R}}{1000^\circ\text{R}} = 0.6 \text{ (60\%)} \leftarrow$$

Also, the net work of this cycle is

$$W_{\text{cycle}} = Q_H - Q_C$$

$$= (Q_H - Q) + (Q - Q_C)$$

$$= 2 W_{\text{cycle}} \leftarrow$$

1. The combined cycle thermal efficiency is greater than either of the individual cycles denoted R1 and R2.

PROBLEM 5.44

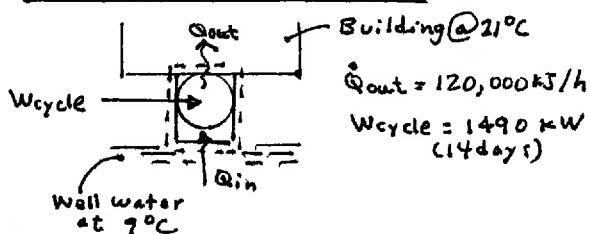
A heat pump cycle is used to maintain the interior of a building at 21°C . At steady state, the heat pump receives energy by heat transfer from well water at 9°C and discharges energy by heat transfer to the building at a rate of $120,000 \text{ kJ/h}$. Over a period of 14 days, an electric meter records that $1490 \text{ kW} \cdot \text{h}$ of electricity is provided to the heat pump. Determine

- the amount of energy that the heat pump receives over the 14-day period from the well water by heat transfer, in kJ.
- the heat pump's coefficient of performance.
- the coefficient of performance of a reversible heat pump cycle operating between hot and cold reservoirs at 21°C and 9°C .

KNOWN: Operating data are provided for a heat pump at steady state that heats a building while receiving energy by heat transfer from well water.

FIND: Determine the amount of energy received from the well water, the heat pump's coefficient of performance, and the coefficient of performance of a reversible heat pump working between the same temperature levels

Schematic & Given Data:



ENGINEERING MODEL:

- The heat pump is the system.
- The building and well water play the roles of hot and cold reservoirs, respectively.
- The system is at steady state.
- All energy transfers are positive in the directions of the arrows.

ANALYSIS: (a) Applying an energy balance using assumption 4,

$$W_{\text{cycle}} = \dot{Q}_{\text{out}} - \dot{Q}_{\text{in}} \Rightarrow \dot{Q}_{\text{in}} = \dot{Q}_{\text{out}} - W_{\text{cycle}}$$

where

$$\dot{Q}_{\text{out}} = (120,000 \frac{\text{kJ}}{\text{h}}) (\frac{24 \text{ h}}{\text{day}}) (14 \text{ days}) = 40.32 \times 10^6 \text{ kJ}$$

$$W_{\text{cycle}} = (1490 \text{ kW} \cdot \text{h}) \left| \frac{3600 \text{ s}}{1 \text{ h}} \right| \left| \frac{1 \text{ kW}}{1 \text{ kW}} \right| = 5.36 \times 10^6 \text{ kJ}$$

$$\Rightarrow \dot{Q}_{\text{in}} = (40.32 - 5.36) \times 10^6 \text{ kJ} = 34.96 \times 10^6 \text{ kJ} \quad \leftarrow \dot{Q}_{\text{in}}$$

(b) The coefficient of performance is

$$\gamma = \frac{\dot{Q}_{\text{out}}}{W_{\text{cycle}}} = \frac{40.32 \times 10^6 \text{ kJ}}{5.36 \times 10^6 \text{ kJ}} = 7.52 \quad \leftarrow \gamma$$

(c) For a reversible heat pump operating between reservoirs at $T_H = 273^\circ\text{C} + 21^\circ\text{C} = 294 \text{ K}$ and $T_C = 273^\circ\text{C} + 9^\circ\text{C} = 282 \text{ K}$,

$$\gamma_{\text{MAX}} = \frac{T_H}{T_H - T_C} = \frac{294 \text{ K}}{12 \text{ K}} = 24.5 \quad \leftarrow \gamma_{\text{max}}$$

PROBLEMS 5.60, 5.61, 5.62

5.60 The figure gives the schematic of a vapor power plant in which water steadily circulates through the four components shown. The water flows through the boiler and condenser at constant pressure and through the turbine and pump adiabatically. Kinetic and potential energy effects can be ignored. Process data follow:

Process 4-1: constant-pressure at 1 MPa from saturated liquid to saturated vapor

Process 2-3: constant-pressure at 20 kPa from $x_2 = 88\%$ to $x_3 = 18\%$

(a) Using Eq. 5.13 expressed on a time-rate basis, determine if the cycle is *internally reversible*, *irreversible*, or *impossible*.

(b) Determine the thermal efficiency using Eq. 5.4 expressed on a time-rate basis and steam table data.

(c) Compare the result of part (b) with the *Carnot efficiency* calculated using Eq. 5.9 with the boiler and condenser temperatures and comment.

SCHEMATIC & GIVEN DATA:

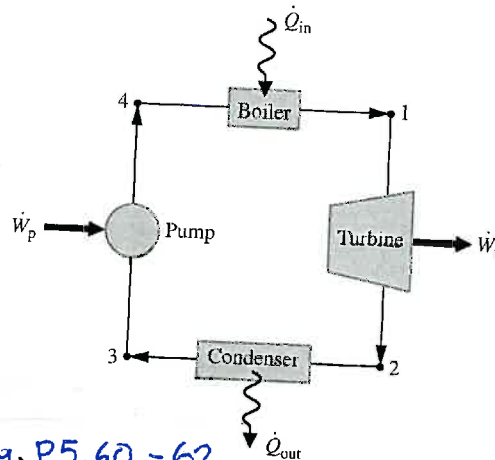
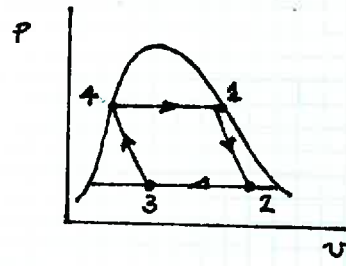


Fig. P5.60-62

ENGINEERING MODEL:

1. A control volume at steady state encloses each of the four components
2. Power and heat transfer rates are positive in the direction of the arrows.
3. Stray heat transfer and kinetic and potential energy effects can be ignored



ANALYSIS:

(a) Expressed on a time-rate basis, Eq. 5.13 takes the form:

$$\dot{Q}_{\text{cycle}} = \frac{\dot{Q}_{23}}{T_{23}} - \frac{\dot{Q}_{41}}{T_{41}}$$

where

$$\dot{Q}_{41} = \dot{m}(h_1 - h_4) = \dot{m}[h_{fg}]_{41}$$

$$\begin{aligned} \dot{Q}_{23} &= \dot{m}[h_2 - h_3] = \dot{m}[(h_f + x_2 h_{fg}) - (h_f + x_3 h_{fg})] \\ &= \dot{m}[(x_2 - x_3)][h_{fg}]_{23} \end{aligned}$$

Collecting results,

$$\frac{\dot{Q}_{\text{cycle}}}{\dot{m}} = \frac{(x_2 - x_3)(h_{fg})_{23}}{T_{23}} - \frac{(h_{fg})_{41}}{T_{41}} \quad (1)$$

With data from Table A-3, $T_{41} = 179.9^\circ\text{C}$, $(h_{fg})_{41} = 2015.3 \text{ kJ/kg}$

$T_{23} = 60.06^\circ\text{C}$, $(h_{fg})_{23} = 2358.3 \text{ kJ/kg}$

$$\frac{\dot{Q}_{\text{cycle}}}{\dot{m}} = \frac{1650.81}{333.21 \text{ K}} - \frac{2015.3 \text{ kJ/kg}}{453.05 \text{ K}} = 0.51 \frac{\text{kJ/kg}}{\text{K}} \quad \text{irreversible}$$

PROBLEMS 5.60, 61, 62 (Continued)

$$(b) \eta = \frac{\dot{W}_{cycle}}{\dot{Q}_{in}} = 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}} = 1 - \frac{(x_2 - x_3)[h_{fg}]_{23}}{[h_{fg}]_{41}} \quad (2)$$

$$\Rightarrow \eta = 1 - \frac{(0.88 - 0.18)(2358.3)}{2015.3} = 0.18 \quad (18\%)$$

$$(c) \eta_{MAX} = 1 - \frac{T_C}{T_H} = 1 - \frac{T_{23}}{T_{41}} \quad (3)$$

$$= 1 - \frac{333.21}{453.05} = 0.265 \quad (26.5\%)$$

Since $\eta < \eta_{MAX}$, the cycle is irreversible.

5.61 Repeat Problem 5.60 for the following case:

Process 4-1: constant pressure at 8 MPa from saturated liquid to saturated vapor

Process 2-3: constant pressure at 8 kPa from $x_2 = 67.5\%$ to $x_3 = 34.2\%$

Data from Table A-3:

$$T_{41} = 295.1^\circ\text{C}, (h_{fg})_{41} = 1441.3 \frac{\text{kJ}}{\text{kg}}$$

$$T_{23} = 41.51^\circ\text{C}, (h_{fg})_{23} = 2403.1 \frac{\text{kJ}}{\text{kg}}$$

$$(a) \text{ Using Eq. (1), } \dot{Q}_{cycle} = \frac{800.23}{\text{in}} - \frac{1441.3}{568.25} = \text{rounded value} \quad \text{internally reversible} \leftarrow$$

$$(b) \text{ Using Eq. (2), } \eta = 1 - \frac{800.23}{1441.3} = 0.445 \quad (44.5\%) \leftarrow$$

$$(c) \text{ Using Eq. (3), } \eta_{MAX} = 1 - \frac{314.66}{568.25} = 0.446 \quad (44.6\%) \leftarrow$$

The results of (b) and (c) agree to within round-off. The cycle is internally reversible.

5.62 Repeat Problem 5.60 for the following case:

Process 4-1: constant pressure at 0.15 MPa from saturated liquid to saturated vapor

Process 2-3: constant pressure at 20 kPa from $x_2 = 90\%$ to $x_3 = 10\%$

Data from Table A-3:

$$T_{41} = 111.4^\circ\text{C}, (h_{fg})_{41} = 2226.5 \frac{\text{kJ}}{\text{kg}}$$

$$T_{23} = 60.06^\circ\text{C}, (h_{fg})_{23} = 2358.3 \frac{\text{kJ}}{\text{kg}}$$

$$(a) \text{ Using Eq. (1), } \dot{Q}_{cycle} = \frac{1886.64}{333.21} - \frac{2226.5}{384.55} = -0.13 \frac{\text{kJ/kg}}{\text{K}} \quad \text{impossible} \leftarrow$$

$$(b) \text{ Using Eq. (2), } \eta = 1 - \frac{1886.64}{2226.5} = 0.153 \quad (15.3\%)$$

$$(c) \text{ Using Eq. (3), } \eta_{MAX} = 1 - \frac{333.21}{384.55} = 0.134 \quad (13.4\%) \leftarrow$$

Since $\eta > \eta_{MAX}$, the cycle is impossible.